Essay 1: crawler access and disclosure-window liquidity

CURRENT PAPER — SHORT UPDATE

Question and design

- Do firm-initiated crawler restrictions change how markets process mandatory filings?
- Treatment: first restriction of any of 31 search/discovery crawlers in archived `robots.txt` files.
- Staggered design, 2021–2025: 27,327 10-K/10-Q events; 1,908 firms.
- Outcome: filing-window log Amihud minus its local benchmark.

Current result

+0.0134

Abnormal log Amihud (BJS), $p = 0.010$

- Benchmark liquidity improves faster; the filing window does not keep pace.
- Quoted spreads: null; dollar volume: up.
- Return variance: up; post-filing drift: null.
- **Reading:** price impact/depth, not trading withdrawal.

Feedback: sharpen the contribution and causal story

COMMITTEE FEEDBACK + OUR DISCUSSION

01

Measurement paper

Candidate contribution: the first systematic evidence that firms **deliberately reduce the machine discoverability of their own public information**.

Validate actual discoverability, not only the stated policy.

02

Mechanism

Amihud is not information deterioration. Separate less information, informed-trader composition, noise-trader exit, and disclosure content.

Adjudicate with spreads, volume, breadth, and price response.

03

More outcome variables

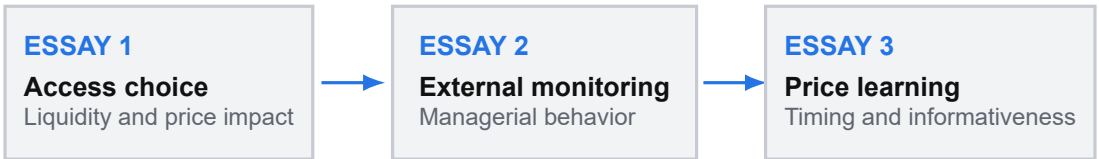
Expand the outcome battery, but require each variable to distinguish a mechanism rather than add another correlation.

Families: price response, investor breadth, monitoring, disclosure.

New architecture: three uses of discoverability

NEW DISSERTATION ARCHITECTURE

Public information must be discoverable before it can discipline managers or enter prices.



Common theoretical primitive

Firms can raise outsiders' cost of discovering and processing legally public information.

Separation across essays

Distinct equilibrium objects; Essay 3 should ideally use separate identifying variation.

Essay 1: Choosing to Become Harder to Find

MEASUREMENT + MARKET MICROSTRUCTURE

Research question

Do firms restrict crawler access to their websites, and does this access choice change trading around mandatory disclosures?

Core evidence

- New monthly panel of crawler-access policies and staggered switches.
- Filing-window price impact relative to local market conditions.

Contribution

- The supply of discoverability is a firm choice, not a fixed feature of public disclosure.
- Information can remain legally public while becoming economically harder to find.
- The access choice has a measurable market-microstructure consequence.

Boundary: this is not an AI paper, and it does not equate illiquidity with informational inefficiency.

Essay 2: External monitoring

TURNING OFF THE MACHINES — ACCESS AND THE EROSION OF MONITORING

Core question

Can a firm weaken outside monitoring by withdrawing machine access to information that remains publicly available?

Mechanism

Machine discovery is an input into scalable monitoring. Blocking raises detection costs and lowers the probability that irregular behavior is noticed.

New object: monitoring technology, not trading liquidity.

Empirical design

- Access restrictions → managerial outcomes; use passive/vendor-driven adoption where possible.
- Heterogeneity by machine-information reliance and conventional monitoring.
- Placebos: routine insider trades; firms with strong institutional monitoring.

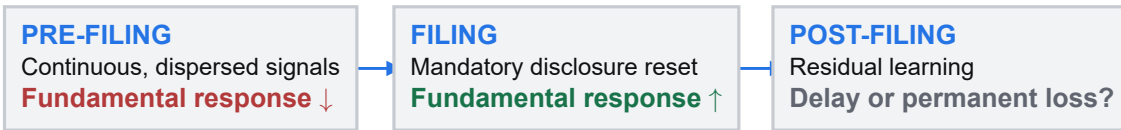
Primary outcome families

- Opportunistic Form 4 trading and trading profitability.
- Discretionary accruals and real earnings management.
- Late filings, amendments, restatements, and reporting discipline.

Where Prices Learn

ESSAY 3 — SEARCH DISCOVERY AND THE TIMING OF FUNDAMENTAL INFORMATION

Does lower discoverability shift learning from the months between filings to the filing date?



Temporal reallocation

Pre-filing response falls, filing response rises, and the full-cycle response is unchanged. Information is delayed, not destroyed.

Information destruction

The filing response does not fully compensate; total price–fundamental covariance falls or post-filing learning persists.

Main requirement: link returns to future fundamentals. Variance alone shows when prices move, not when they learn.

When is Essay 3 a separate paper?

OVERLAP AUDIT AGAINST ESSAY 1

MERGE INTO ESSAY 1 IF

- It uses the same robots event, sample, filing window, and staggered event study.
- Price delay, variance ratios, synchronicity, or announcement volatility are simply added as outcomes.
- The conclusion is only that restrictions worsen market quality.

STANDALONE IF

- The estimand is the **timing versus total amount** of fundamental learning.
- The unit is the inter-filing learning cycle, not filing-window illiquidity.
- Returns are anchored to future earnings or another external fundamental realization.
- Preferably, identification comes from a separate external shock to search discovery.

70–80%

Estimated design overlap in this weak version

Strongest version: ask when prices learn, not whether trading becomes less liquid.

Three essays, one coherent dissertation

DISSERTATION MAP

	Essay 1	Essay 2	Essay 3
Question	Do firms restrict discoverability, and does liquidity change?	Does removing machine access weaken external monitoring?	Does search discovery change when prices learn?
Core estimand	Filing-window price impact relative to local liquidity	Change in opportunistic managerial behavior	Reallocation of fundamental response across time
Preferred variation	Firm crawler-access restrictions	Restrictions plus passive/vendor-driven adoption variation	Separate external search-discovery shock; robots as fallback
Literature home	Market microstructure	Corporate governance and disclosure	Information economics and asset pricing

Common spine: firms shape effective access to public information. Distinct objects: liquidity, governance, and informativeness.

Questions for the meeting

DECISIONS THAT DETERMINE THE NEXT ITERATION

1. **Is "firms choose to become less discoverable" strong enough as Essay 1's measurement contribution?**
2. **Should Essay 2 prioritize governance/monitoring over analyst information production?**
3. **For Essay 3, should I preserve the robots link or require a separate discoverability shock?**

Next two weeks

- Run a direct discoverability first-stage pilot.
- Lock the Essay 1 mechanism horse race and outcome battery.
- Build minimum-viability pilots for Form 4 outcomes and a separate discovery shock.

Goal for the meeting: choose the dissertation architecture before expanding the empirical workload.